Can imaging see innate immune memory? A public-data audit with a self-supervised video model of innate-cell behaviour

Rafid Quadir *rafidquadir@gmail.com*

Abstract

Trained immunity and tolerance change how monocytes and macrophages respond to a second stimulus, but they are measured by destructive restimulation assays. Whether these memory states leave a signature that label-free imaging could read has not been tested, because no public dataset images trained or tolerised cells. We asked how far public data can take the question. We trained a self-supervised video model of innate-cell behaviour on 279,647 clips of migrating human neutrophils and tested it, with strong image and video baselines, on every public dataset that pairs innate-cell imaging or live-cell dynamics with a molecular, metabolic, signalling or functional readout, summarised as a modality coverage map. Imaging features separate microglia activation morphologies, and NF-κB reporter dynamics encode stimulus identity, but frozen image features did not predict cell identity or NAD(P)H lifetime beyond simple intensity statistics, and a general-purpose video model outperformed the cell-specific one. Memory states themselves are paired with imaging in no public resource. From the restimulation transcriptomes that do exist, after sample-identity checks, we find that β-glucan-trained human macrophages have lower TNF mRNA than untrained cells 4 h after restimulation in two independent datasets (six of six donors; -2.09 and -1.96 log2), although the same studies report higher TNF-α protein at 24 h. A single readout time can therefore show training as either enhanced or reduced, which shapes the imaging study we propose to test innate immune memory directly.

1 Introduction

Innate immune cells keep a memory of earlier stimuli. Monocytes and macrophages exposed to β-glucan or BCG respond more strongly to an unrelated second stimulus days later (trained immunity), while cells exposed to LPS respond less (tolerance); both states are carried by epigenetic and metabolic reprogramming rather than by antigen-specific receptors (Netea et al., 2016; Saeed et al., 2014; Novakovic et al., 2016). These states are measured today by restimulating cells and assaying cytokines, transcripts or chromatin, which destroys the sample and gives one time point. If memory states left a signature in how living cells look and move, label-free video could read them out continuously and non-destructively.

Self-supervised video models make that question testable without labels: a joint-embedding predictive architecture (JEPA) learns representations by predicting the latent content of masked or future parts of a clip (Bardes et al., 2024; Assran et al., 2025). But testing whether such a model captures immune state requires paired data, images or video of cells whose molecular or functional state was measured, and for trained innate cells none is public. This paper asks how far one can get with public data only. We train TiJepa, a video JEPA, on 279,647 unlabelled clips of migrating human neutrophils, and test its frozen representations, against strong image and video models, on every public dataset we could find that pairs label-free images, video or image-derived dynamics of innate immune cells with a molecular, functional or memory readout.

The contribution is an audit rather than a new state of the art, and it has three parts.

1. A *modality coverage map* (Figure 1, Table 1): for each modality family (immunophenotype, metabolism, signalling, transcriptomics, epigenomics, cytokines, integrated omics), the best pub-

lic pairing with imaging, the level at which it is paired (same cell, same condition, different studies), what we tested and the result, including the families for which no public pairing exists.

2. A controlled evaluation of what a video JEPA learns from innate-immune video: a matched-step spatial-only ablation, a frozen general-purpose video model (V-JEPA 2), image models, random-initialisation and frame-shuffle controls, a label-free validation set used for every model choice, controls that separate forecasting from positional structure in the latent space, and a diagnosis of the partial dimensional collapse we observed.

3. From the molecular data that exist without images, a replicated observation that constrains what an imaging study of trained cells should measure, and the design of that study.

Most of our answers are negative or null, and we report them as such. No result in this paper shows trained immunity in images: no public image of trained cells exists, and the memory-related analyses use the closest public proxies (acute priming and tolerance, polarisation, restimulation transcriptomes).

2 Modality coverage map

Table 1 and Figure 1 summarise, for each modality family, the best public pairing of innate immune cell imaging with that modality and what it supports. Three things stand out. First, same-cell pairings exist only for immunophenotype (morphology classes, cell type), metabolism (NAD(P)H lifetime) and signalling (reporter dynamics), and none of them involves a memory state; the one memory-adjacent same-cell resource, sequential NF-κB stimulation, has two wells per arm. Second, transcriptome and chromatin data are paired with dynamics only at the level of five conditions, which supports description, not prediction. Third, the families in which innate memory is actually measured, cytokines after restimulation and the transcriptional and epigenetic programmes behind them, have no public imaging pairing at all; the public restimulation transcriptomes (Section 3) are what an imaging study of memory would have to be validated against. Proteomics beyond signalling reporters, integrated single-cell multi-omics, spatial omics, perturbation screens, repertoire, microbiome and clinical data have no public pairing with in-vitro innate cell imaging. For the same reason we did not project image features onto molecular activation axes: no public dataset has both images and a stimulus-matched molecular anchor.

Within the same-cell pairings, frozen image or video features carry information about morphology classes and NF-κB dynamics carry information about stimulus identity, but where a simple measured quantity is available as a reference (autofluorescence intensity for cell identity; the photon-count image for lifetime), the learned features did not add to it.

3 What restimulation transcriptomes say that imaging would need to detect

No public dataset images trained or tolerised monocytes, but three public datasets measure their transcripts after restimulation. We use them to ask which differences an imaging study would have to reproduce, with three safeguards: every restimulation label is checked against the GEO sample records and against a positive control (restimulated untrained cells must exceed unstimulated untrained cells for TNF, IL6 and IL1B in every donor), every expected direction is quoted from the dataset's source publication, and sample identity is verified before any observation is reported.

Sourced directions. Of the 7 TNF and IL-6 directions the source publications report after restimulation, 5 agree in sign with the mean donor-paired difference in the deposited transcripts: dimethyl itaconate (GSE212282) and IL-4 (GSE185433) increase TNF and IL6 transcripts, as their source papers report for the secreted proteins (Ferreira et al., 2023; Schrijver et al., 2023). The two exceptions are both β-glucan.

β-glucan lowers the TNF transcript at 4 h after restimulation, in two independent datasets. In GSE212282, β-glucan-trained macrophages restimulated with LPS for 4 h on day 6 have less TNF mRNA than untrained cells in all three donors (log2 difference -2.2 to -2.0), although the source paper reports more

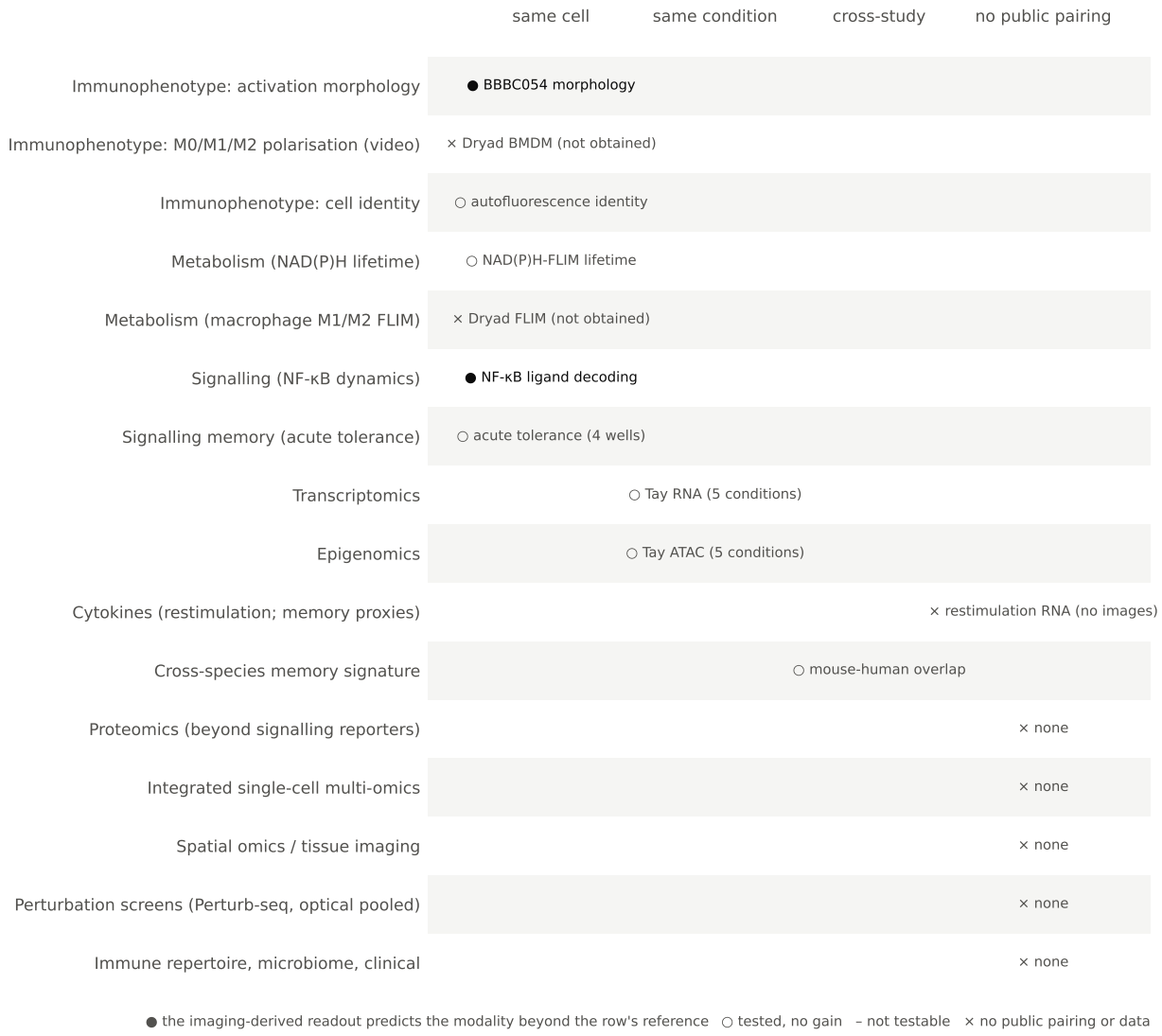


Figure 1: Modality coverage map. Each row is a modality family; the column is the level at which the best public dataset pairs it with images, video or image-derived dynamics of innate immune cells. Filled circle: frozen image or dynamics features carry information beyond the simple baseline of that row; open circle: tested, no such gain; cross: no public pairing or data not obtainable. Details, numbers and sources in Table 1.

secreted TNF-α after 24 h of restimulation (Ferreira et al., 2023). Before reporting this we checked sample identity. All 75 samples agree on sex within each donor; the three donors carry distinct presence/absence genotypes at HLA-DRB5 and GSTM1, consistent across every sample of a donor, so no sample belongs to another donor; and after removing each donor's mean, every β-glucan sample's profile is most similar to the other donors' β-glucan samples (day-6 restimulated arms: 14/15 samples matched to their own arm, the one exception being a β-glucan + DMI sample matched to DMI). Over the protocol the β-glucan effect on TNF reverses: it raises the transcript during the first exposure (4 h: +2.0 to +3.0; day 1: +2.4 to +3.1), is inconsistent at rest on day 6 (-1.6 to +0.7) and lowers it after restimulation (Figure 2).

The observation replicates in GSE141656, an independent dataset none of whose three donors matches a GSE212282 donor at all four genotype-like loci we compared: after 4 h of heat-killed *M. tuberculosis* on day 6, β-glucan-trained macrophages have less TNF mRNA in every donor (-2.07 to -1.83; restimulation positive

Table 1: Modality coverage map: the best public pairing of each modality family with images, video or image-derived dynamics of innate immune cells, what was tested and the result. BA: balanced accuracy.

Modality family	Pairing	Public data	Test and result	Outcome
Immunophenotype: activation morphology	same cell	BBBC054 (mouse microglia, LPS time course; CC BY 3.0)	3-class morphology from frozen features, FOV-grouped: best frozen encoder V-JEPA 2 ViT-L BA 0.681 vs hand-crafted 0.499; TiJepa-1 0.595	positive
Immunophenotype: M0/M1/M2 polarisation (video)	same cell	Dryad m0cfxpp8v (BMDM movies)	polarisation from frozen video features: not obtained: the download requires an interactive browser check	absent
Immunophenotype: cell identity	same cell	Zenodo 15657646 (two-photon autofluorescence, T cells and neutrophils in the same fields; CC BY 4.0)	T cell vs neutrophil, FOV-grouped, mixed fields only: best frozen encoder TiJepa-0 BA 0.721; intensity-only baseline 0.714 (CI 0.663-0.760)	null
Metabolism (NAD(P)H lifetime)	same cell	Zenodo 14330511 (two-photon NAD(P)H-FLIM, macrophages; CC BY 4.0)	per-cell mean lifetime from the photon-count image, FOV-grouped ridge: best frozen encoder TiJepa-1 (spatial-only) R^2 +0.32; intensity baseline +0.12; knowing only the experimental arm +0.50; within fields: TiJepa-1 +0.08, intensity +0.08	null
Metabolism (macrophage M1/M2 FLIM)	same cell	Dryad 8gtht76q2	not obtained: the download requires an interactive browser check	absent
Signalling (NF-κB dynamics)	same cell	Tay lab deposit, Zenodo 14337874 (live-cell NF-κB reporter, 125 wells)	11-way ligand-pair identity from single-cell traces, well-grouped: trace JEPA BA 0.286, raw trace 0.269, hand-crafted 0.198; reference: chance 0.091. Pretraining adds nothing over the same network at random initialisation	positive
Signalling memory (acute tolerance)	same cell	Tay deposit, LPS then LPS (4 wells)	tolerant vs naive from the second response; power analysis: unpowered (2 wells per arm); 8 wells per arm needed for 80% power at the observed effect	null
Transcriptomics	same condition	Tay RNA-seq GSE260996 (same lab as the imaging; 5 conditions)	condition-level dynamics $\rightarrow$ RNA (Q^2 vs leave-one-out mean): fraction of top-2,000 genes with $Q^2 > 0$: 0.226 (exact permutation p 0.36; 5 conditions, descriptive)	null
Epigenomics	same condition	Tay ATAC-seq GSE261647 (5 conditions)	condition-level dynamics $\rightarrow$ accessibility: fraction of top-2,000 peaks with $Q^2 > 0$: 0.182 (exact permutation p 0.44; 5 conditions, descriptive)	null
Cytokines (restimulation; memory proxies)	no public pairing	GSE212282, GSE185433, GSE141656 (human macrophage training and tolerance, RNA only; no images)	TNF/IL6 transcripts after restimulation, donor-paired: β-glucan-trained cells: TNF transcript lower at 4 h after LPS (3/3 donors) and after M. tuberculosis (3/3, independent donors)	absent
Cross-species memory signature	cross-study	Tay mouse RNA (priming x LPS) vs GSE185433 human tolerance	overlap of priming-blunted LPS genes among genes induced in both species: overlap 18 vs 24.5 expected, OR 0.58, two-sided p 0.09	null
Proteomics (beyond signalling reporters)	no public pairing	-	no public dataset pairs images of innate immune cells with proteomics	absent
Integrated single-cell multi-omics	no public pairing	-	no public dataset pairs live-cell imaging of innate cells with single-cell multi-omics of the same cells	absent
Spatial omics / tissue imaging	no public pairing	-	in-tissue data exist but not for the in-vitro cell systems studied here	absent
Perturbation screens (Perturb-seq, optical pooled)	no public pairing	-	no public optical or Perturb-seq screen of trained or tolerised innate cells	absent
Immune repertoire, microbiome, clinical	no public pairing	-	not applicable to in-vitro innate-cell imaging	absent

control at least 7.9 log2 units in every donor), and IL6 (-0.43) and IL1B (-0.80) move the same way. Its source publication, too, reports more TNF-α protein after 24 h of restimulation (Moorlag et al., 2020). A third dataset with only two replicates (GSE242947, Cheng et al., 2024) is consistent at 1 h of LPS (-1.49, -0.76) and inconsistent at 3 h (+0.41, -0.51).

We therefore report a replicated, transcript-level observation: at 4 h after restimulation, β-glucan-trained human macrophages have lower TNF mRNA than untrained ones, in two datasets and six donors, while the same studies report higher TNF-α protein at 24 h. One explanation is kinetic, that trained cells reach their

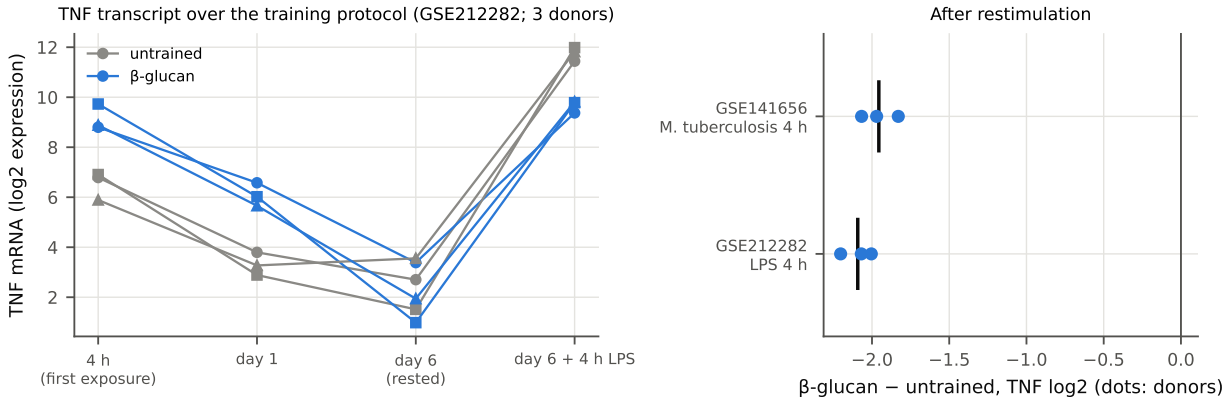


Figure 2: TNF transcripts in β-glucan-trained human macrophages. Left: TNF mRNA in untrained (grey) and β-glucan-trained (blue) cells of three donors across the training protocol of GSE212282; marker shape identifies the donor. Right: trained-minus-untrained TNF after 4 h of restimulation in two independent datasets; dots are donors, bars means.

TNF transcript peak earlier and resolve it sooner; it is a hypothesis, and the test is a restimulation time course from 0.5 to 24 h with paired RNA and protein. With three donors per dataset no confidence interval is reported; the smallest achievable exact p for a three-donor sign-flip test is 0.25, so the evidence is the consistency across six of six donors and two restimuli, not a p-value.

4 Signalling dynamics: what live-cell traces encode

The Tay-lab deposit (Wang et al., 2025) is the one public resource in which live-cell dynamics of macrophages, here single-cell NF-κB reporter traces, are paired at the condition level with transcriptome and chromatin data, and it includes sequential stimulations, which is the structure innate memory has.

Ligand-pair identity is decodable from dynamics. Across 125 wells and 32,048 cells, the identity of the ordered pair of stimuli (11 classes, chance 0.091) is decoded from single-cell traces well above chance, with wells as groups (Figure 3): the raw trace reaches 0.269 (95% CI 0.239–0.285), hand-crafted trace features 0.198 (0.172–0.217), and a trace JEPA, pretrained without labels on 118,243 traces from 495 wells that are not evaluated, 0.286 (0.263–0.305). The pretraining is not what helps: the same network at random initialisation reaches 0.284 (difference +0.003, 95% CI -0.008–+0.027), and its advantage over the raw trace (+0.020, +0.002–+0.043) does not survive correction for multiple comparisons. Hand-crafted summaries lose information that the full trace carries (+0.083, +0.067–+0.107).

Acute tolerance cannot be tested with the deposited wells. The deposit has two wells of LPS-then-LPS and two of naive-then-LPS cells, so a well-level test of tolerance from dynamics has no power. From the observed well means of the second response's peak amplitude (standardised effect -1.58) and their between-well SD, a two-sided test at $\alpha = 0.05$ needs 8 wells per arm for 80% power, and 30 if the SD is at the upper limit of its 80% interval; for the response's area under the curve the numbers are 7 and 24. Because that SD rests on four wells, we read this as a lower bound on what a tolerance-from-dynamics study needs, not as a precise design.

Dynamics to omics and across species. With five matched conditions, condition- level dynamics do not predict held-out transcript or accessibility profiles better than the leave-one-out mean (Table 1), and among 343 orthologs induced by LPS in both mouse macrophages and human macrophages, the genes blunted by heterologous priming in mouse overlap the genes blunted by homologous tolerance in human no more than expected by chance (18 observed, 24.5 expected, odds ratio 0.58, two-sided $p = 0.09$). Both analyses are exploratory.

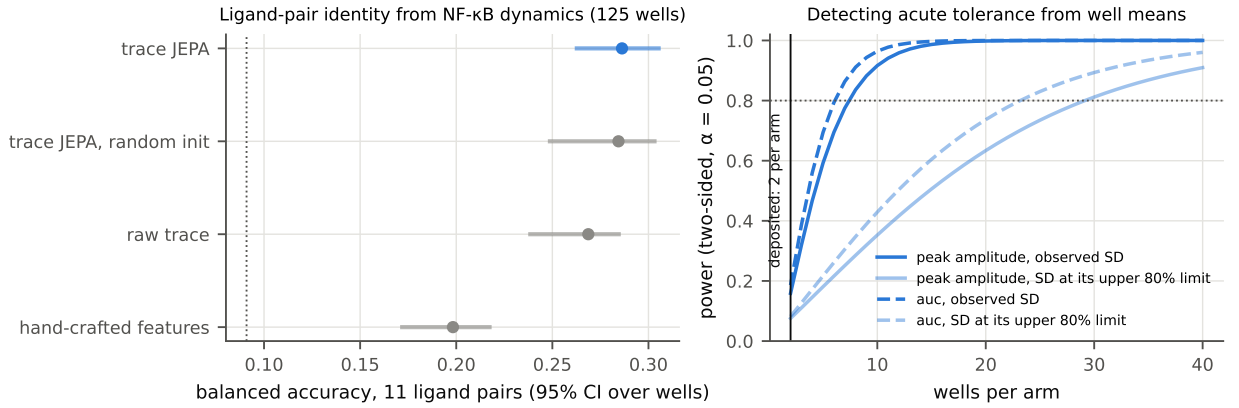


Figure 3: NF-κB dynamics. Left: decoding the ordered ligand pair from single-cell traces (95% CI over wells; dotted line: chance). Right: power of a well-level test for acute tolerance as a function of wells per arm, at the observed effect (solid) and with the between-well SD at the upper limit of its 80% interval (faint).

5 What a video JEPA learns from innate-immune video

5.1 Morphology of activated microglia

On BBBC054, every learned representation separates the three microglia morphology classes far better than hand-crafted shape and texture features (0.499) or exposure time alone (0.401; Figure 4). The pretrained video model V-JEPA 2 is the best encoder (0.681). TiJepa-1 (0.595) is not distinguishable from DINOv2 (0.610), the ImageNet ViT (0.590) or TiJepa-0 (0.607); V-JEPA 2 exceeds it by 0.086 (95% CI 0.067–0.108). Training on cell video therefore did not beat a general-purpose video model, and twice as many training steps on a quality-filtered corpus did not improve on TiJepa-0. Within the TiJepa recipe, future masking adds +0.013 (+0.001–+0.027) over the spatial-only ablation and training adds +0.041 (+0.027–+0.055) over a randomly initialised network of the same architecture, which itself reaches 0.554. In the round-versus-amoeboid task the ordering is the same at a higher level (V-JEPA 2 0.860, TiJepa-1 0.818, DINOv2 0.775). Accuracy depends on apparent cell size: rescaling the crops by 0.5, 0.75, 1 and 1.5 before encoding gives 0.516, 0.545, 0.588 and 0.616 for TiJepa-1, so differences in magnification between pretraining and evaluation data matter.

5.2 Label-free cell identity

On T cells and neutrophils imaged in the same fields, frozen encoders reach balanced accuracies of 0.681–0.721 on the 20 mixed fields, every one of them inside the 95% CI of a probe on per-channel intensity statistics (0.714, CI 0.663–0.760; Figure 4). TiJepa-1 reaches 0.714 and V-JEPA 2 0.718; random initialisation reaches 0.696. On the gradient-contrast channel alone, which is closest to the brightfield TiJepa was trained on, TiJepa-1 gives 0.655 and V-JEPA 2 0.696. Cell identity in these data is carried by autofluorescence intensity; frozen image features add nothing measurable to it. In the six-class set, where each cell type was imaged in its own session, the hand-crafted features reach 0.755 and TiJepa-1 0.650; that comparison cannot separate cell type from session.

5.3 NAD(P)H lifetime from the photon image

In 1,657 macrophages from 50 fields, the per-cell mean NAD(P)H lifetime (1.95 ± 0.13 ns; our data check required a median between 0.3 and 3.5 ns) is predicted from frozen features of the photon-count image with R^2 up to +0.32 (TiJepa-1 +0.30, V-JEPA 2 +0.31), against +0.12 for intensity statistics. That gain does not reflect cell-level metabolism. Lifetime differs between experimental arms, and a reference that predicts each cell by its arm's training-fold mean reaches $R^2 = +0.50$, more than any image feature. When targets and

features are centred on their field's mean, so that only variation between cells of one field counts, TiJepa-1 reaches +0.08 (95% CI +0.04–+0.11), V-JEPA 2 +0.04 and intensity statistics +0.08 (+0.04–+0.11); the bound fraction behaves the same way. These two controls were added after seeing the first result. The frozen features recognise the condition a field came from; they do not read a cell's lifetime beyond its brightness.

5.4 Forecasting is explained by positional structure

TiJepa-1's predictor forecasts its own future latents with an error 0.17 times that of copying the last context latent, against 0.85 for the spatial-only ablation, which never saw a future mask, and the advantage holds at every stride (Figure 5). It also beats a slot-conditional mean fitted on other movies (ratio 0.32, exact $p = 0.031$, the smallest possible with six movies), so its predictions are specific to the clip. But the controls show that the advantage over copy-last is not evidence of modelling cell dynamics. On static clips, in which nothing moves, the predictor keeps 0.79 of its 0.83 improvement over copy-last; given the context of a different clip from the same movie it keeps 0.51. Copy-last is a weak reference because the target encoder's latents change with temporal position even when the image does not, and the predictor receives those positions. Decoded to pixels, the predictions are no closer to the future frames than the last observed frame (error ratio 1.41 at stride 1, 1.06 at stride 27). Our pre-specified criterion for forecasting fails, and we make no forecasting claim.

5.5 Re-identifying cells over time

On the 64 held-out validation movies, re-identifying the same cell after 3 to 27 frames among all cells at that time (chance 0.078) separates the encoders clearly (Figure 4). V-JEPA 2 is best (0.667, 95% CI 0.638–0.695), followed by the ImageNet ViT (0.614) and DINOv2 (0.585); raw downsampled pixels reach 0.555 (0.522–0.589). TiJepa-1 reaches 0.501 (0.470–0.533), below raw pixels. Among TiJepa models the training objective matters: the spatial-only ablation reaches only 0.298 (0.273–0.327), and randomly initialised networks 0.212–0.228. Shuffling the frames of TiJepa-1's input clips lowers its score to 0.470, so it uses the order of frames only slightly. Because this set is used to compare training recipes, these numbers describe the models; they are not a held-out test.

5.6 Associating cells across frames

With ground-truth cells in the six held-out movies, centroid distance alone matches 0.883 of cells correctly across a 9-frame gap (IDF1 over the movie 0.625; Figure 5). Embeddings alone are worse for every encoder: V-JEPA 2 0.751, TiJepa-1 0.404, DINOv2 0.411, random initialisation 0.208. Adding the embedding to position lowers accuracy for every encoder except V-JEPA 2, whose combination (0.875) is not distinguishable from position alone (exact $p = 0.250$); for TiJepa-1 it falls to 0.723 ($p = 0.031$, the smallest possible with six movies). Within the TiJepa family the ordering of the re-identification task recurs: on embedding distance alone TiJepa-1 exceeds its frame-shuffled input by 0.089 and the spatial-only ablation by 0.151 (both $p = 0.031$), and V-JEPA 2 exceeds TiJepa-1 by 0.347. For densely packed migrating neutrophils at an 8 s frame interval, position is the identity cue; appearance features of any model we tested add nothing to it.

5.7 Dimensional collapse

The effective rank of TiJepa-1's pooled embedding on a fixed set of microglia crops fell from 15.8 at step 14,751 to 8.1 at the end of training, and its per-dimension standard deviation from 0.25 to 0.04; the spatial-only ablation ended at 7.6 (Figure 6). The collapse is a property of the encoder, not of the probe input: on held-out validation clips the pooled target-encoder and context-encoder embeddings of TiJepa-1 have the same effective rank (7.7 and 7.7 of 384), the first principal component carries 0.49 of the variance and the first eight 0.87. Individual tokens are less collapsed (effective rank 59.4), so the information lost is mostly in what survives mean pooling. TiJepa-0's pooled embedding has rank 29.6 on the same clips. Across the six pretrained and random networks, effective rank on the validation clips is correlated with both the microglia accuracy (Spearman $\rho = +0.88$) and the re-identification score ($\rho = +0.77$), which is descriptive with six models. In the training logs the rank of the training-batch embedding peaks within the first 20,000 steps

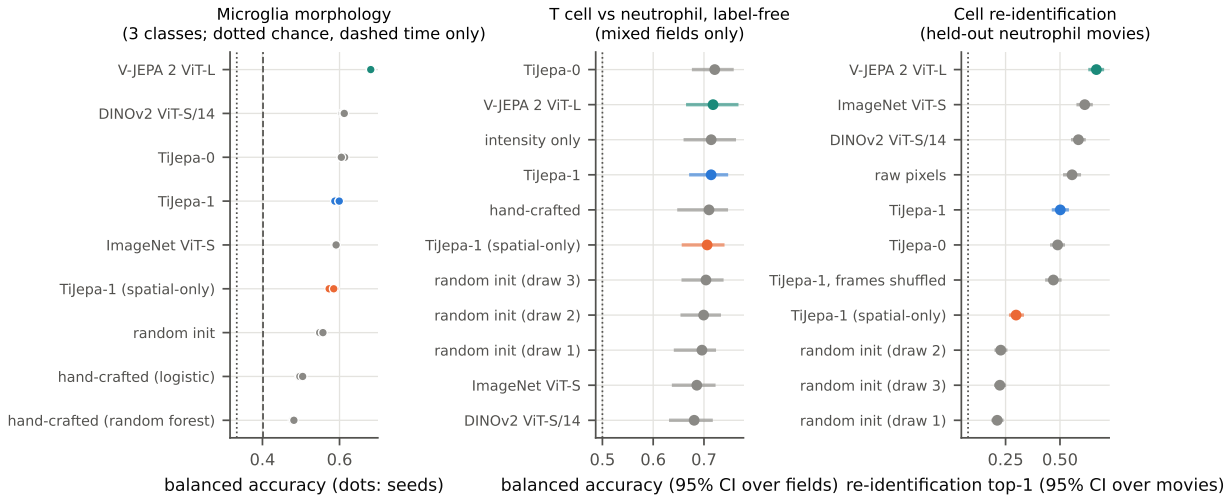


Figure 4: Frozen-feature benchmarks. Left: microglia morphology (BBBC054), three-class balanced accuracy, one dot per probe seed. Middle: T cells versus neutrophils from label-free autofluorescence, on fields that contain both (95% CI over fields). Right: re-identification of the same neutrophil after 3–27 frames on held-out movies (95% CI over movies); this label-free validation set is the only data used to compare training recipes.

and declines while the EMA momentum rises towards 1 and the learning rate decays, so the cause cannot be separated from the schedule without changing the recipe; Section 5.8 reports a short recipe study that changes it.

5.8 A short recipe study

To ask whether a change of recipe prevents the collapse, we trained five variants from scratch for 20,000 steps each (schedules ending at 20,000 steps, snapshots every 5,000) and scored every snapshot only on the validation set: the TiJepa-1 recipe (S0), EMA momentum capped at 0.9995 (S1), a variance–covariance penalty on the predictor outputs (Bardes et al., 2021) (S2), an effective batch of 256 with a linearly scaled learning rate (S3), and future targets drawn from long-stride clips with a deeper predictor (S4). Before training we fixed the criterion for adopting a variant: a validation score at least 0.02 above S0 and at least twice its effective rank, both at 20,000 steps. No variant met it. S0 reached 0.359 with effective rank 45.2; the variants changed the score by -0.011 to +0.018 and the rank by a factor of 0.98 to 1.31 (Figure 6). At 20,000 steps none of the runs had yet collapsed (rank 44.5–59.4, against 7.7 for TiJepa-1 after 117,000 steps), so this budget cannot test the late-training collapse; it shows only that none of these changes improves the early representation enough to justify a full retraining.

6 Discussion

What the video JEPA learned. Trained on 279,647 clips of one cell type from one instrument, TiJepa-1 learned representations that classify microglia morphology far better than hand-crafted features. Image models trained on millions of natural images do as well, and a general-purpose video model used frozen and off the shelf, V-JEPA 2, does better. On the label-free validation task TiJepa-1 is below raw pixels. Future masking helps relative to the matched spatial-only ablation, modestly on morphology and strongly on re-identification, but the predictor's apparent forecasting skill is explained by the positional structure of its own target latents: it persists on clips in which nothing moves. The representation is partially collapsed, and rank and downstream performance move together across models. A practitioner who wants frozen features of innate-cell video today should start from a large pretrained video model.

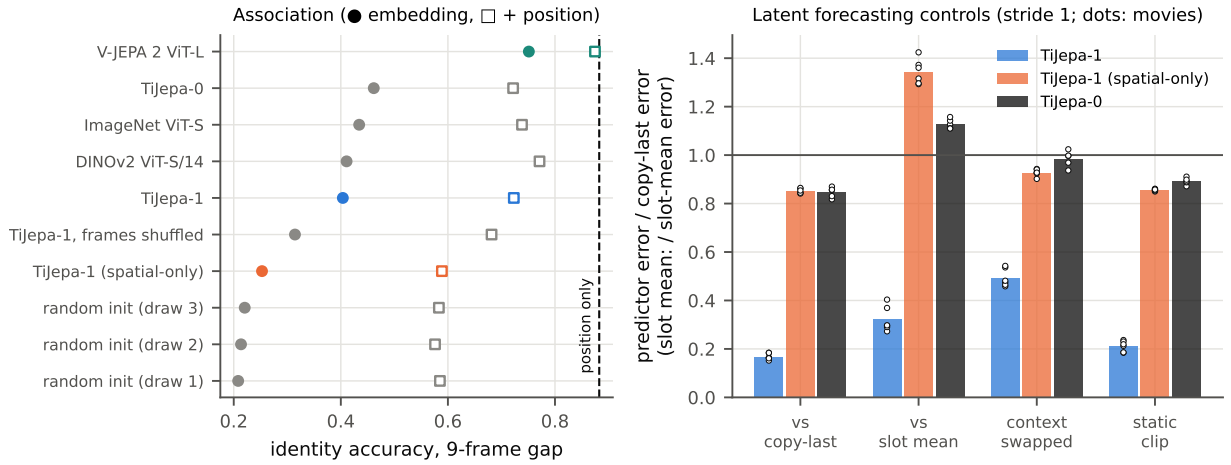


Figure 5: Association and forecasting. Left: identity matching of ground-truth cells across a 9-frame gap from embedding distance alone (filled) and combined with centroid distance (open); dashed line: centroid distance alone. Right: latent forecasting error relative to copying the last context latent (first, third and fourth groups) or to the slot-conditional mean fitted on other movies (second group), at stride 1; dots are movies. A forecasting model should lose its advantage under a swapped context and on static clips; TiJepa-1 keeps most of it.

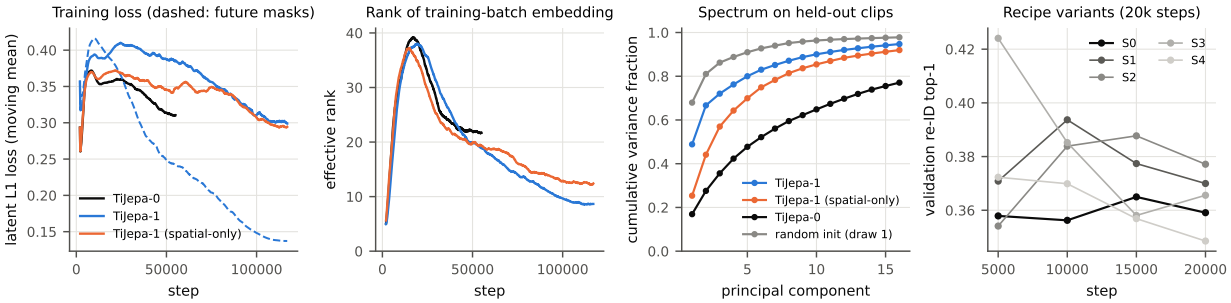


Figure 6: Training and dimensional collapse. Left: latent loss of TiJepa-0, TiJepa-1 and the spatial-only ablation (solid: spatially masked batches; dashed: future-masked batches). Second: effective rank of the pooled target embedding of training batches (64 clips). Third: cumulative variance of the leading principal components of the pooled target embedding on held-out validation clips. Right: validation re-identification score of the five 20,000-step recipe variants at each snapshot.

What public data can and cannot answer. The coverage map makes the central gap concrete: every public pairing of innate-cell imaging with a molecular or functional readout is either at the same-cell level for states that are not memory (morphology classes, cell type, metabolic lifetime, stimulus identity), or at the condition level with five conditions. Where a simple measured reference exists (autofluorescence intensity for cell identity, the photon image for lifetime) the learned features did not add to it. The memory states themselves are measured only by restimulation assays that have never been imaged.

A constraint for the experiment that is missing. The replicated observation that β-glucan-trained human macrophages have less TNF transcript than untrained cells 4 h after restimulation, in two datasets with different restimuli and donors, while the same studies report more TNF-α protein at 24 h, means that a single readout time can show training as either enhanced or reduced. An imaging study of trained immunity therefore needs its molecular ground truth at more than one time after restimulation, and its imaging window should cover the hours in which the transcript and protein responses diverge (Section 6.1).

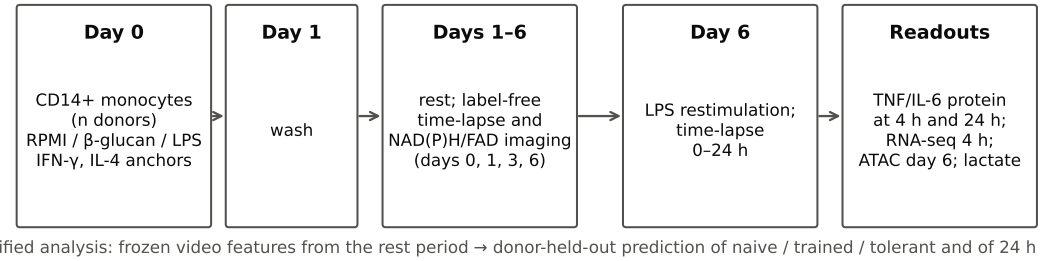


Figure 7: The proposed imaging study of trained and tolerised human monocytes.

6.1 Future work: the experiment that would test this directly

No public dataset images trained or tolerised human monocytes, so whether innate immune memory has an imaging phenotype remains untested. The study that would test it (Figure 7) uses primary human $CD14^+$ monocytes from independent donors in five arms: medium, β-glucan (training), LPS (tolerance), and IFN-γ and IL-4 as polarisation anchors with known morphological and metabolic effects. Cells are exposed for 24 h, washed and rested for 5 days, then restimulated with LPS on day 6. Label-free brightfield time-lapse is recorded during the rest and for 0–24 h after restimulation, at a frame interval chosen to cover the horizons our forecasting analysis spanned (8 s to 3.6 min); NAD(P)H/FAD autofluorescence, with lifetime imaging where available, is acquired on days 0, 1, 3 and 6. Ground truth is TNF and IL-6 protein at 4 h and 24 h after restimulation, bulk RNA-seq at 4 h, ATAC-seq on day 6 and lactate in the supernatant.

The analysis is fixed in advance: frozen video features (TiJepa, DINOv2 and V-JEPA 2) of the rest-period movies, donor-held-out classification of naive, trained and tolerised wells, and donor-held-out prediction of 24 h TNF protein from pre-restimulation video, with image-free baselines (well-level intensity and cell counts) as the reference; the primary endpoint is the gain over that reference. The hypothesis is falsified if no frozen feature set exceeds the reference with a donor-level confidence interval excluding zero. Effects between donors are consistent in the transcript data (the SD of the trained-minus-untrained TNF difference after restimulation is 0.10 and 0.12 log2 units in the two datasets), but nothing constrains the size of an imaging effect. Assuming a standardised donor-level effect of 1.0 (0.8, 1.5), a paired two-sided test at $\alpha = 0.05$ needs 10 (15, 6) donors for 80% power. Images would be deposited in the BioImage Archive under CC BY, sequencing in GEO, and cytokine and plate metadata alongside.

7 Limitations

Cell types. The model is pretrained on human neutrophil migration, evaluated on neutrophils, mouse microglia, mouse immune cells and FLIM-imaged macrophages, and the memory biology concerns human monocyte-derived macrophages. Every transfer across that chain is an assumption the data cannot check; the activation-labelled macrophage movies that would have tested it directly could not be obtained.

Seeds and scale. One pretraining seed was run per recipe on one GPU, so differences between pretrained models carry no seed-to-seed uncertainty; the probe seeds reported vary only the folds and subsamples. TiJepa is a ViT-S trained for about 11.9 GPU-hours on a single-source corpus; V-JEPA 2 is a ViT-L trained on orders of magnitude more video.

Collapse. Feature effective rank fell throughout training for every recipe we trained to completion (Section 5.7); our 20,000-step recipe study (Section 5.8) ends before the collapse sets in and does not establish its cause.

Small groups. Six tracking movies, four tolerance wells, five matched omics conditions and three donors per restimulation dataset are below the size at which we report confidence intervals; those analyses give per-group values and exact tests, and the tolerance and dynamics-to-omics analyses are descriptive.

Readouts. The restimulation observation is at the transcript level at 4 h; the 24 h protein results it contrasts with are those reported by the source publications, not re-measured. The NAD(P)H lifetime readouts are computed without deconvolving the instrument response and are relative measures.

8 Related work

Self-supervised video representation learning. JEPA-style video models predict masked spatio-temporal regions in latent space rather than pixels (Bardes et al., 2024); V-JEPA 2 scales the approach to over a million hours of video and adds action-conditioned prediction (Assran et al., 2025). Image self-supervision (DINOv2, Oquab et al., 2023) and supervised ImageNet ViTs (Steiner et al., 2021) are the usual frozen baselines. Variance and covariance regularisation counteracts dimensional collapse in joint-embedding training (Bardes et al., 2021).

Label-free imaging of immune cell state. Label-free two-photon autofluorescence of NAD(P)H and FAD has been used to classify immune cell types (Kreiss et al., 2026), and NAD(P)H lifetime imaging reports the metabolic state of macrophages (Musimbi et al., 2025). High-throughput migration imaging of neutrophils in multi-well plates is available at scale (Cibir et al., 2023). Live-cell NF-κB reporter dynamics of macrophages exposed to sequential ligand pairs have been deposited together with transcriptome and chromatin measurements of the same conditions (Wang et al., 2025).

Innate immune memory. Trained immunity and tolerance are defined functionally by the response to restimulation (Netea et al., 2016) and are accompanied by epigenetic and transcriptional reprogramming of monocytes and macrophages (Saeed et al., 2014; Novakovic et al., 2016). Public restimulation datasets measure transcripts and, in their source publications, secreted cytokines (Ferreira et al., 2023; Schrijver et al., 2023; Moorlag et al., 2020); none includes images.

9 Methods

9.1 Pretraining data

The pretraining corpus consists of label-free brightfield time-lapse movies of human neutrophils migrating in multi-well plates, from the ComplexEye deposit (Cibir et al., 2023) (Zenodo 10082674, CC BY 4.0). Ten of the deposit's 21 archives were used. Every movie was checked for duplication against the held-out tracking movies before use (Section 9.6), passed a per-frame focus and illumination check, was kept at its native pixel size (0.641 µm/px, within 1.5% of the 0.65 µm/px target), and was cut into cell-centred clips of 16 frames of 96×96 pixels that follow a tracked cell. Clips were taken at temporal strides of 1, 3, 9 and 27 frames (frame interval 8 s, so a clip spans 2 min to 60 min), and 10% of clips were replaced by a single frame repeated 16 times so that the model also sees static input. The frozen corpus holds 279,647 clips from 407 movies; its index and shards are fixed by checksum, and every model below was trained on exactly this set.

9.2 Model

TiJepa is a video joint-embedding predictive architecture (Bardes et al., 2024). A ViT-S encoder (12 layers, width 384, 6 heads) embeds $2 \times 8 \times 8$ tubelets of a $16 \times 96 \times 96$ single-channel clip into an $8 \times 12 \times 12$ token grid with fixed 3-D sine-cosine positional embeddings. A context encoder sees only the visible tokens; a target encoder, an exponential moving average (EMA) of the context encoder, sees the full clip; a 6-layer predictor of width 192 receives the context tokens and the positional embeddings of the masked positions and predicts the target encoder's layer-normalised tokens there, with an L1 loss. Each batch uses one of two masks: with probability 0.5 a *future* mask that hides every spatial position of the last two temporal slots

(the last 4 frames), otherwise a spatial tube mask of 8 random blocks spanning all frames. The *spatial-only* ablation uses only the second mask. Representations for every downstream task are the mean-pooled tokens of the frozen target encoder.

9.3 Training

Both TiJepa-1 and its spatial-only ablation were trained from scratch for 117,000 steps of batch 128 (53.5 passes over the corpus) with AdamW (learning rate 1.5×10^{-4} after 500 warm-up steps, cosine decay to 0, weight decay 0.05, $\beta = (0.9, 0.95)$), bf16 autocast and gradient clipping at 1.0; the EMA momentum rose linearly from 0.996 to 1.0. Both schedules end exactly on the last step. TiJepa-1 used 11.9 and the ablation 9.2 A100-40GB GPU-hours. TiJepa-1 was trained in two segments, resumed from a checkpoint with its data stream, masks and optimiser state restored. We also report TiJepa-0, an earlier model trained for 55,000 steps with spatial masking only on a different corpus (7 archives, 288,081 clips, no quality filter); it is a historical reference, not an ablation, because its corpus differs. One pretraining seed was run per recipe.

9.4 Frozen baselines

All encoders are frozen; only linear probes are fitted. Baselines are an ImageNet-trained ViT-S/16 (Steiner et al., 2021), DINOv2 ViT-S/14 (Oquab et al., 2023), the general-purpose video model V-JEPA 2 ViT-L (Assran et al., 2025) (MIT licence; Hugging Face `facebook/vjepa2-vitl-fpc64-256` at a pinned revision), the same TiJepa architecture at random initialisation (three independent draws), TiJepa-1 with the frames of each clip shuffled, and dataset-specific hand-crafted features. Grayscale input is replicated to three channels for image and video models and resized to each model's native resolution with its own normalisation; still images are repeated over 16 frames for video models. Image models see the last frame of a clip.

9.5 Evaluation protocol

Every probe is fitted with group-held-out cross-validation, where the group is the unit of acquisition (field of view, movie, well or donor), so no group contributes to both the training and the test folds. Unless stated otherwise the probe is standardisation, PCA to 64 dimensions and class-balanced logistic regression (ridge regression for continuous targets), with three seeds that vary the fold assignment and, where the data are subsampled, the subsample. We report balanced accuracy (BA) or R^2. Confidence intervals are 95% cluster bootstraps over groups (2,000 resamples) and are reported only with at least 10 groups; comparisons between encoders are paired bootstraps on the same resamples with Benjamini–Hochberg correction within a task. With fewer than 10 groups we report every group's value and exact sign-flip or permutation tests, whose smallest achievable p is stated. Model and recipe choices were made only on a label-free validation set that is never used as a test set; the scores reported on it describe the models and are not held-out results (Section 9.7); every reported number comes from a model frozen before its test set was used. Every result records the checkpoint (file, step, SHA-256) it came from, and every model name in this paper is derived from that record.

9.6 Leakage control

Every corpus movie was fingerprinted (block-averaged frames at the start, middle and end) and compared with the six held-out tracking movies; candidate pairs were re-tested at full resolution. One corpus movie duplicated a held-out movie and was excluded from the corpus; TiJepa-0's older corpus contained it, so that movie is dropped from every TiJepa-0 tracking summary. The validation movies were checked against both the held-out movies and the corpus (Section 9.7).

9.7 Benchmarks

Validation: cell re-identification. Sixty-four movies from a ComplexEye archive that is not in the corpus form the validation set. Cells were detected and linked with the same classical pipeline used to build the corpus. At four anchor frames per movie, every cell tracked without gaps from 15 frames before to 42

frames after the anchor is embedded from the 16-frame clip ending at the anchor (the query) and from the clip starting $k \in \{3, 9, 27\}$ frames later (the gallery; query and gallery share no frame). The score is top-1 cosine retrieval of the same cell among all cells at that anchor, averaged over k; chance is one over the number of cells. Its leakage check searched three cell patches from every corpus movie in the validation movie at the same time index (no match above normalised cross-correlation 0.78, against 0.84–1.00 for a patch searched in its own movie) and fingerprinted it against the held-out movies.

Microglia morphology (BBBC054). About 8,775 annotated mouse microglia per seed from BBBC054 (Ljosa et al., 2012) (CC BY 3.0), sampled with a per-(field, class) cap, are classified as ramified, round or amoeboid from a 96×96 crop, with 60 fields of view as groups. The dataset is an LPS exposure time course, so we also report a probe given only the exposure time.

Label-free immune cell identity. Two-photon autofluorescence patches (two band-pass channels for NAD(P)H and FAD, plus a Dodt gradient-contrast channel; 64×64 pixels, 25.6 µm) of mouse immune cells (Kreiss et al., 2026) (Zenodo 15657646, CC BY 4.0). In the primary set, CD4 T cells and neutrophils were imaged in the same fields and labelled by an anti-CD3 channel that no model sees; we classify them with fields as groups, and separately on the 20 fields that contain both types, which removes any field-level shortcut. In a secondary six-class set each type was imaged in its own samples, so cell type and session are confounded. Encoders embed each channel separately and the three embeddings are concatenated. The reference is a probe on per-channel intensity statistics.

NAD(P)H lifetime. Raw two-photon time-correlated single-photon counting data of macrophages (Musimbi et al., 2025) (Zenodo 14330511, CC BY 4.0): 227 lifetime bins of 0.055 ns over 512×512 pixels, five repeats per field, summed. Cells were segmented on the photon image; per cell we computed the first moment of the background-subtracted decay tail from 0.5 ns after the peak (mean lifetime) and the bound fraction from a non-negative fit of the tail to $e^{-t/0.4\,\mathrm{ns}}$, $e^{-t/3.4\,\mathrm{ns}}$ and a constant (free and enzyme-bound NAD(P)H). Encoders see only the photon-count (intensity) crop. The task is ridge regression of each cell's lifetime readouts with fields as groups.

Detection-level association. On the six S-BIAD2196 movies of migrating neutrophils with ground-truth instance masks (BioImage Archive accession S-BIAD2196), every annotated cell in frames t and $t + k$ ($k \in \{1, 3, 9\}$) is embedded from the 16-frame clip ending at that frame in a fixed window centred on the cell, and identities are matched by the Hungarian algorithm on embedding cosine distance, on centroid distance, or on both (each cost matrix z-scored and summed with equal weight). We report the fraction of cells present in both frames that are matched to themselves, and IDF1 over a chain of 9-frame associations through each movie.

Latent forecasting and its controls. On cell-centred clips from the same six movies at strides 1, 3, 9 and 27, each model's predictor is given the first 12 frames and predicts its target encoder's latents for the last 4, through the future mask. Its L1 error is compared, within the model's own latent space, with: copying the last context latent; the mean target latent at each future slot (and slot and position), fitted on the other half of the movies; the predictor given the context of a different clip from the same movie (context swap); and the predictor on a clip made of one frame repeated 16 times (static). A ridge decoder from target latents to the future patches' pixels, fitted on the other half of the movies, checks the predictions in pixel space. Before running the controls we fixed the criterion for calling this forecasting: the predictor must beat both copy-last and the slot mean (exact sign-flip $p < 0.05$ over movies) and must lose at least half of its advantage over copy-last under both the context swap and the static control.

9.8 Statistics software

Analyses used Python with PyTorch, scikit-learn, SciPy, statsmodels and pydeseq2 (Muzellec et al., 2023). Moderated paired tests for expression data follow the empirical-Bayes approach of Smyth (2004).

Data and code availability

Every dataset is public and was downloaded directly from its repository; files were checked against the repository's MD5 checksums where it publishes them, and the SHA-256 of every file used is recorded with the code. Table 2 lists accession, licence and use. One planned dataset, Dryad `doi:10.5061/dryad.m0cfxpp8v` (bone-marrow macrophage movies with M0/M1/M2 labels, CC0), could not be obtained because its download requires an interactive browser check; the macrophage FLIM set `doi:10.5061/dryad.8gtht76q2` was not obtained for the same reason. Model weights: V-JEPA 2 ViT-L (MIT licence), DINOv2 ViT-S/14 and the AugReg ImageNet ViT-S/16 (Apache-2.0). Code, configuration and the analysis records behind every number are in the project repository, which will be linked here when it is made public.

Table 2: Datasets used. GEO records carry no licence; their originating publications are cited.

Dataset	Accession	Licence	Use
ComplexEye neutrophil migration movies (Cibir et al., 2023)	Zenodo 10082674	CC BY 4.0	pretraining (10 archives); validation set (one further archive)
CellTrack neutrophil movies with instance masks	BioImage Archive S-BIAD2196	CC BY 4.0	association, forecasting
Microglia LPS time course (Ljosa et al., 2012)	BBBC054	CC BY 3.0	morphology probe
Label-free autofluorescence of immune cells (Kreiss et al., 2026)	Zenodo 15657646	CC BY 4.0	cell identity
NAD(P)H-FLIM raw data (Musimbi et al., 2025)	Zenodo 14330511	CC BY 4.0	lifetime prediction
NF-κB live-cell traces (Wang et al., 2025)	Zenodo 14337874	CC BY 4.0	ligand decoding, tolerance power
Matched RNA-seq and ATAC-seq (Wang et al., 2025)	GEO GSE260996, GSE261647	GEO	dynamics to omics; mouse side of the cross-species test
Training and restimulation RNA-seq (Ferreira et al., 2023)	GEO GSE212282	GEO	restimulation, sample QC
Tolerance and IL-4 RNA-seq (Schrijver et al., 2023)	GEO GSE185433	GEO	restimulation, human side of the cross-species test
β-glucan and *M. tuberculosis* RNA-seq (Moorlag et al., 2020)	GEO GSE141656	GEO	independent replication
Dectin-1 ligand training RNA-seq (Cheng et al., 2024)	GEO GSE242947	GEO	descriptive (2 replicates)
Mouse–human orthologs (Baldarelli et al., 2024)	MGI ortholog table	MGI terms	cross-species test

References

Mido Assran et al. V-JEPA 2: Self-Supervised Video Models Enable Understanding, Prediction and Planning. arXiv:2506.09985, 2025.

Richard M Baldarelli et al. Mouse Genome Informatics: an integrated knowledgebase system for the laboratory mouse. *GENETICS*, 2024. doi: 10.1093/genetics/iyae031.

Adrien Bardes, Jean Ponce, and Yann LeCun. VICReg: Variance-Invariance-Covariance Regularization for Self-Supervised Learning. arXiv:2105.04906, 2021.

Adrien Bardes et al. Revisiting Feature Prediction for Learning Visual Representations from Video. arXiv:2404.08471, 2024.

Quen J. Cheng et al. Dectin-1 ligands produce distinct training phenotypes in human monocytes through differential activation of signaling networks. *Scientific Reports*, 2024. doi: 10.1038/s41598-024-51620-8.

Zülal Cibir et al. ComplexEye: a multi-lens array microscope for high-throughput embedded immune cell migration analysis. *Nature Communications*, 2023. doi: 10.1038/s41467-023-43765-3.

Anaísa V. Ferreira et al. Dimethyl itaconate induces long-term innate immune responses and confers protection against infection. *Cell Reports*, 2023. doi: 10.1016/j.celrep.2023.112658.

Lucas Kreiss et al. Cell-MICS: Detecting Immune Cells With Label-Free Two-Photon Autofluorescence and Deep Learning. *Journal of Biophotonics*, 2026. doi: 10.1002/jbio.70260.

Vebjorn Ljosa et al. Annotated high-throughput microscopy image sets for validation. *Nature Methods*, 2012. doi: 10.1038/nmeth.2083.

Simone J.C.F.M. Moorlag et al. β-Glucan Induces Protective Trained Immunity against Mycobacterium tuberculosis Infection: A Key Role for IL-1. *Cell Reports*, 2020. doi: 10.1016/j.celrep.2020.107634.

Zaneta D. Musimbi et al. Two-photon NAD(P)H-FLIM reveals unperturbed energy metabolism of Ascaris suum larvae, in contrast to host macrophages upon artemisinin derivatives exposure. *Scientific Reports*, 2025. doi: 10.1038/s41598-025-85780-y.

Boris Muzellec et al. PyDESeq2: a python package for bulk RNA-seq differential expression analysis. *Bioinformatics*, 2023. doi: 10.1093/bioinformatics/btad547.

Mihai G. Netea et al. Trained immunity: A program of innate immune memory in health and disease. *Science*, 2016. doi: 10.1126/science.aaf1098.

Boris Novakovic et al. β-Glucan Reverses the Epigenetic State of LPS-Induced Immunological Tolerance. *Cell*, 2016. doi: 10.1016/j.cell.2016.09.034.

Maxime Oquab et al. DINOv2: Learning Robust Visual Features without Supervision. arXiv:2304.07193, 2023.

Sadia Saeed et al. Epigenetic programming of monocyte-to-macrophage differentiation and trained innate immunity. *Science*, 2014. doi: 10.1126/science.1251086.

David P. Schrijver et al. Resolving sepsis-induced immunoparalysis via trained immunity by targeting interleukin-4 to myeloid cells. *Nature Biomedical Engineering*, 2023. doi: 10.1038/s41551-023-01050-0.

Gordon K Smyth. Linear Models and Empirical Bayes Methods for Assessing Differential Expression in Microarray Experiments. *Statistical Applications in Genetics and Molecular Biology*, 2004. doi: 10.2202/1544-6115.1027.

Andreas Steiner, Alexander Kolesnikov, Xiaohua Zhai, Ross Wightman, Jakob Uszkoreit, and Lucas Beyer. How to train your ViT? Data, Augmentation, and Regularization in Vision Transformers. arXiv:2106.10270, 2021.

Andrew G. Wang et al. Macrophage memory emerges from coordinated transcription factor and chromatin dynamics. *Cell Systems*, 2025. doi: 10.1016/j.cels.2025.101171.

A Additional details

Hand-crafted image features. For BBBC054 and the autofluorescence and lifetime crops, cells were segmented by a classical pipeline (Otsu threshold on edge energy and local variance, hole filling, distance-transform watershed, the component nearest the centre); features are shape descriptors, Zernike moments, Haralick texture and intensity statistics. A learned segmenter was not used.

Point tracking. An earlier tracking benchmark followed a cell by patch correlation of each encoder's features between consecutive frames. It cannot distinguish representations: TiJepa-1 with shuffled frames, TiJepa-1 and randomly initialised networks all lost the cell after a median of one to two frames, against twenty for a nearest-detection tracker. We report it only to explain why the association benchmark replaces it.

Microglia benchmark. Crops are 96×96 pixels around each annotated cell at the native resolution of the images, which state a $20\times$ objective but no pixel size, so no rescaling to the pretraining pixel size was possible; the scale sensitivity in Section 5 is the consequence. Accuracy computed within each LPS exposure time bin is close to the pooled accuracy (TiJepa-1: 0.609 within bins), so exposure time does not explain the morphology result.

Validation set. The 64 validation movies come from an archive of the pretraining source that was not used to build the corpus. Their leakage check is described in Section 9.7; at a threshold below every self-match (0.80) no duplicate was found either.

Lifetime readouts. Without the instrument response, the decay tail from 0.5 ns after the peak is used; the first moment of a truncated multi-exponential tail and the two-component fit with fixed lifetimes are relative measures that are comparable between cells of one instrument and protocol, which is all the regression requires.

Restimulation datasets. For GSE212282 and GSE185433 the GEO restimulation labels were cross-checked against each sample's characteristics, and all three donors of each pass the restimulation positive control. Expected directions were taken only where the source publication reports TNF or IL-6 after restimulation for that arm; arms without such a statement are not scored. GSE212282's arm labelled DMI is dimethyl itaconate, as in its source paper.

Cross-species test. Mouse: an interaction model of priming (none, CpG, polyI:C) by challenge (none, LPS) on the deposited bone-marrow macrophage RNA-seq (three replicates per condition; pydeseq2); genes LPS-induced in naive cells (adjusted $p < 0.05$, log2 fold change > 1) whose induction is reduced by priming (interaction adjusted $p < 0.05$). Human: the same rule on GSE185433, with untrained unstimulated, untrained restimulated, tolerised unstimulated and tolerised restimulated samples of three donors and donor-paired moderated tests (Smyth, 2004). The universe is one-to-one MGI orthologs LPS-induced in both species (Baldarelli et al., 2024).